# ECON 5140: Applied Econometrics

## Homework 4: IV, Regression Discontinuity & Difference-in-Differences

**Covers:** Lesson 10 (Regression Discontinuity & Instrumental Variables) & Lesson 11 (Difference-in-Differences)

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# PART 1: ANALYTICAL PROBLEMS

## Problem 1: Instrumental variables — assumptions and diagnostics

Consider a binary instrument $Z$, binary treatment $D$, and outcome $Y$. Unobserved confounders $U$ may affect both $D$ and $Y$.

**Questions:**

a) State the three main IV assumptions (relevance, independence/exogeneity, exclusion restriction) in words.

**Answer**: **Relevance** Cov($Z$, $D$) ≠ 0, instrument $Z$ must meaningfully affect treatment $D$; **Independence**: $Z \perp U$, Instrument $Z$ is "as good as randomly assigned" and uncorrelated with unobserved confounders $U$; **Exclusion** The instrument $Z$ affects outcome $Y$ only through $D$, not directly

b) Which of these can you assess directly from data using a single cross-section? How?

**Answer**: **Relevance** Check via first-stage regression $D = \pi_0 + \pi_1 Z + \epsilon$; Test whether $\pi_1 \neq 0$; Look at first-stage F-statistic

c) What is the **Stock–Yogo style** rule of thumb for a *single* instrument in the first stage? What goes wrong when the instrument is weak?

**Answer**: First-stage F-statistic < 10 → weak IV concern, meaning instrument explains little variation in treatment and finite-sample bias likely large. IV estimates become biased; Estimates become unstable; Inference becomes invalid.

## Problem 2: Wald estimator — hand calculation (extended sample)

You observe $n = 16$ units (binary instrument $Z$, binary treatment $D$, outcome $Y$):

| Unit | $Z$ | $D$ | $Y$ | Unit | $Z$ | $D$ | $Y$ |
|---|---|---|---|---|---|---|---|
| 1 | 1 | 1 | 12 | 9 | 1 | 0 | 10 |
| 2 | 1 | 0 | 10 | 10 | 0 | 0 | 6 |
| 3 | 0 | 0 | 6 | 11 | 1 | 1 | 12 |
| 4 | 1 | 1 | 11 | 12 | 0 | 1 | 10 |
| 5 | 0 | 1 | 11 | 13 | 1 | 1 | 12 |

| Unit | $Z$ | $D$ | $Y$ | Unit | $Z$ | $D$ | $Y$ |
|---|---|---|---|---|---|---|---|
| 6 | 0 | 0 | 5 | 14 | 0 | 0 | 5 |
| 7 | 1 | 1 | 11 | 15 | 1 | 0 | 10 |
| 8 | 0 | 0 | 6 | 16 | 0 | 0 | 7 |

**Questions:**

a) Compute the **first stage** $\mathbb{E}[D \mid Z=1] - \mathbb{E}[D \mid Z=0]$ (use the sample counts: eight units have $Z=1$ and eight have $Z=0$). Give the answer as a **reduced fraction** and as a decimal.

**Answer** $\mathbb{E}[D \mid Z=1] - \mathbb{E}[D \mid Z=0] = \frac{5}{8} - \frac{1}{4} = \frac{3}{8} = 0.375$, showing how much Z moves treatment D.

b) Compute the **reduced form** $\mathbb{E}[Y \mid Z=1] - \mathbb{E}[Y \mid Z=0]$ using the **sample means** of $Y$ in each $Z$ group. (First compute $\sum Y$ for $Z=1$ and for $Z=0$.)

**Answer** $\mathbb{E}[Y \mid Z=1] - \mathbb{E}[Y \mid Z=0] = 11 - 7 = 4$, meaning how much Z moves outcomes Y.

c) Compute the **Wald IV estimator** (reduced form ÷ first stage). Give the exact value as a **fraction** and approximate to two decimals.

**Answer** $Wald = \frac{\mathbb{E}[D|Z=1]-\mathbb{E}[D|Z=0]}{\mathbb{E}[Y|Z=1]-\mathbb{E}[Y|Z=0]} = \frac{4}{\frac{3}{8}} = \frac{32}{3} \approx 10.67$

d) Compute the **naive difference in means** $\mathbb{E}[Y \mid D=1] - \mathbb{E}[Y \mid D=0]$ (pool all units with $D=1$ vs $D=0$). Why can this differ substantially from the Wald estimate even when the IV assumptions hold?

**Answer** $\mathbb{E}[Y \mid D=1] - \mathbb{E}[Y \mid D=0] = \frac{79}{7} - \frac{65}{9} = \frac{256}{63} \approx 4.06.$

Because the naive difference compares all treated and untreated units directly, so it can mix in **selection bias / endogeneity**. Treated and untreated units may differ for reasons other than treatment. By contrast, the Wald estimator uses only the exogenous variation induced by Z and identifies a local causal effect for compliers, not the overall treated-vs-untreated mean difference.

e) Interpret the Wald estimate as a **local** causal effect: what subpopulation does it correspond to under **monotonicity** ($D(1) \geq D(0)$ for all units)?

**Answer** Using monotonicity($D(1) \geq D(0)$, the Wald estimator identifies the causal effect for **compliers**, here the Wald estimate is the **LATE** (the local average treatment effect for compliers).

f) Under monotonicity, **always-takers** satisfy $D(0) = D(1) = 1$ and **never-takers** satisfy $D(0) = D(1) = 0$. Using only the **observed** $(Z, D)$ pairs in the table, **list the unit ID(s)** that must be always-takers and the unit ID(s) that must be never-takers. (No proof required—state the mapping from $(Z, D)$ to type.)

**Answer** If we observe Z=0 and D=1, then the unit took treatment even without encouragement, so under monotonicity it must be an always-taker. Then **always taker** should be **Unit5 and Unit12**. Similarly, we can derive **never taker** should be **unit 2, 9, 12**

## Problem 3: Regression discontinuity — bins, thresholds, and causal claims

A marketplace promotes sellers to **"featured"** status if their 30-day quality score $X$ is **at or above 60** (sharp rule: $D_i = 1\{X_i \geq 60\}$). You do not observe seller-level data; you only have **average weekly orders** $Y$ by **score bin** (bins are half-open: $[a, b)$ means $a \leq X < b$, except the top bin includes the right endpoint).

| Score bin $X$ | Bin midpoint | Average orders $Y$ (bin mean) |
|---|---|---|
| $[40, 50)$ | 45 | 28.0 |
| $[50, 60)$ | 55 | 34.0 |
| $[60, 70)$ | 65 | 45.0 |
| $[70, 80]$ | 75 | 50.0 |

**Questions:**

a) In one sentence, write the sharp RD assignment rule linking $D_i$, $X_i$, and the **policy cutoff** $c = 60$ .

**Answer** $D_i = 1\{X_i \geq 60\})$

b) A simple **binned RD** comparison at the policy threshold uses the bin **just below** vs **just above** $c$. Compute:$\hat{\tau}_{60} = \bar{Y}_{[60,70)} - \bar{Y}_{[50,60)}$.

Interpret $\hat{\tau}_{60}$ as an approximate **local** treatment effect **at** the cutoff (what group is it informative about?).

**Answer** $\hat{\tau}_{60} = \bar{Y}_{[60,70)} - \bar{Y}_{[50,60)} = 45 - 34 = 11$. This is an approximate local treatment effect for sellers near the cutoff X=60.

c) **Placebo / alternative thresholds.** Suppose someone ignores the actual rule and computes the same type of **across-bin gap** at cutoffs where **treatment does not change** under the true policy (still $D = 1$ only if $X \geq 60$). Compute:

- $\hat{\tau}_{50} = \bar{Y}_{[50,60)} - \bar{Y}_{[40,50)} = 34 - 28 = 6$
- $\hat{\tau}_{70} = \bar{Y}_{[70,80]} - \bar{Y}_{[60,70)} = 50 - 45 = 5$

d) **Causal inference:** Which of $\hat{\tau}_{50}$, $\hat{\tau}_{60}$, $\hat{\tau}_{70}$ is a candidate for a **causal** effect of featured status on orders **under the RD assumptions at the stated cutoff**? Explain why the other two gaps are **not** interpretable as the causal effect of the program at those $X$ values, even though they are "differences in bin averages." (Hint: treatment status is **constant** on both sides of a placebo split, or the policy rule is not at that $X$.)

**Answer** $\hat{\tau}_{60}$ is a candidate causal effect under RD, because treatment changes at X=60, $\hat{\tau}_{50}$, $\hat{\tau}_{70}$ are not causal treatment effects since treatment status does not change at those thresholds.

## Problem 4: Difference-in-differences (DID)

A pilot program ran in City A (treated) but not City B (control). Average sales ($Y$) in thousands:

| | Pre | Post |
|---|---|---|
| **City A (treated)** | 20 | 28 |
| **City B (control)** | 18 | 21 |

**Questions:**

a) Compute the **naive** post-minus-pre change for City A only. Why is this not a convincing causal estimate of the program effect?

**Answer** 28- 20 = 8, Because City A may have changed over time even without the program (common trends, seasonality, macro shocks, or other time-varying factors). So the **before-after change alone does not isolate the treatment effect**.

b) Compute the **difference-in-differences** estimate using the table above.

**Answer** Treated change = 28 - 20 =8, controled change = 21 - 18 = 3, DID= (28– 20) –(21– 18)= 8–3= 5.

c) In the canonical 2 $\times$ 2 DID, what estimand is typically targeted (ATE, ATT, or something else)?

**Answer** DID identifies **ATT** because it compares the observed outcomes of treated units with their counterfactual outcomes (constructed using the control group), but does not observe what would happen if control units were treated.

d) Explain **parallel trends** in plain English. With only **one** pre-treatment period and **one** post period, can you *test* parallel trends statistically? Why or why not?

**Answer** Parallel trends means that,Treated and control groups can differ in levels, but they must evolve in parallel over time (without treatment).

With only one pre-treatment period and one post-treatment period, this assumption cannot be statistically tested because there is no pre-treatment trend to examine.

e) A saturated regression with an interaction,

$$Y_{it} = \beta_0 + \beta_1 \,\mathrm{Treated}_i + \beta_2 \,\mathrm{Post}_t + \beta_3 \left(\mathrm{Treated}_i \times \mathrm{Post}_t\right) + \varepsilon_{it},$$

recovers the DID estimand as $\hat{\beta}_3$ in the 2 $\times$ 2 case **if** the model is correctly specified. In one sentence: does "running TWFE" *create* identification, or only **compute** a DID estimand under a design assumption you must defend separately?

**Answer** Running TWFE does not create identification; it only computes the DID estimand under a design assumption, such as parallel trends, that must be justified separately.

jupyter nbconvert --execute --to html HM4_noncode.ipynb --output noncode_HW4